

SSL Handshake Protocol - Important Questions & Answers

Q1. Why is SSL/TLS needed?

SSL/TLS is needed to provide secure communication over the Internet by ensuring confidentiality, integrity, and authentication.

Q2. Why does SSL use TCP?

SSL uses TCP because TCP provides reliable, ordered, and error-checked communication.

Q3. Explain the purpose of the SSL Handshake Protocol.

The SSL Handshake Protocol authenticates communicating parties, negotiates algorithms, and securely generates cryptographic keys.

Q4. Explain the four phases of SSL Handshake Protocol.

1. Establish Security Capabilities
2. Server Authentication and Key Exchange
3. Client Authentication and Key Exchange
4. Finish Phase

Q5. Why does the client send ClientHello?

ClientHello initiates the handshake and sends supported cipher suites, SSL versions, and random values.

Q6. Explain the role of ServerHello.

ServerHello selects the SSL version, cipher suite, and sends the server random value.

Q7. Why are random values exchanged in SSL handshake?

Random values help generate unique session keys and prevent replay attacks.

Q8. Why does the server send a certificate?

The server sends a certificate to prove its identity and provide its public key.

Q9. Explain how SSL prevents man-in-the-middle attacks.

SSL prevents MITM attacks through certificate verification and authenticated key exchange.

Q10. Explain RSA key exchange in SSL.

The client encrypts the pre_master_secret using the server's public key. The server decrypts it using its private key.

Q11. Explain Diffie-Hellman key exchange in SSL.

Both client and server exchange public values and independently compute the same shared secret.

Q12. Why is the pre_master_secret encrypted?

It is encrypted so attackers cannot read it during transmission.

Q13. Why is the master secret important?

The master secret is used to generate encryption keys, MAC keys, and IVs.

Q14. Explain how the master secret is created.

Both client and server use the pre_master_secret along with client and server random values to generate the master secret.

Q15. Why are separate client and server keys generated?

Separate keys improve security and allow secure communication in both directions.

Q16. Explain the role of ChangeCipherSpec message.

It informs the other side to begin using negotiated encryption and MAC algorithms.

Q17. Why is the Finished message critical?

It verifies the integrity of all handshake messages and confirms secure setup.

Q18. Explain how SSL provides confidentiality.

SSL uses symmetric encryption with shared secret keys to protect data.

Q19. Explain how SSL provides integrity.

SSL uses MACs to detect message modification.

Q20. Explain SSL session and SSL connection.

An SSL session stores security parameters and may support multiple SSL connections. A connection transfers encrypted data.

Q21. Why does SSL use both symmetric and asymmetric encryption?

Asymmetric encryption securely exchanges keys, while symmetric encryption efficiently protects bulk data.

Q22. Why does SSL generate new keys for each session?

New session keys improve security and reduce the impact of compromised keys.

Q23. Explain the steps of SSL Record Protocol.

1. Fragmentation
2. Compression
3. MAC generation
4. Encryption
5. Header addition
6. Transmission

Q24. Why is MAC added before encryption?

MAC ensures message integrity and detects tampering.

Q25. Why is the Finished message encrypted?

Encryption confirms both parties successfully generated the same session keys.